

DARRELL RAFA ALAMSYAH

Full Stack Developer & AI/ML Integrator

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Showcase of 5 selected projects submitted for the Apple Developer Academy Indonesia. Each project highlights interest & motivation, creativity, interdisciplinary potential, and technical problem-solving capabilities.

PROJECT #1: E-Learning Platform

Tech Stack: Next.js, Appwrite, Groq API | **Origin:** Coursework (Software Engineering)

Team: Group Project (4 Members) | **My Role:** Lead Fullstack Developer & AI Integrator

Project Summary

An interactive e-learning platform that gamifies school coursework into a progression-based adventure for children, featuring a virtual economy and an on-demand AI tutor to personalize study guides.

My Impact & Contribution

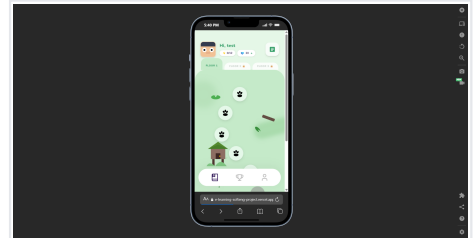
Engineered the core progression database models, built the virtual reward economy, and integrated the Groq API prompt pipeline to analyze student quiz failures and generate adaptive explanation nodes.

Key Learnings

Mastered Next.js App Router server component architecture, complex client-side state synchronizations, and how to construct context-rich system prompts for predictable AI responses.

Reference Links

GitHub: github.com/darrellrafa/e-learning | Live: e-learning-softeng-project.vercel.app



PROJECT #2: Roadmap-AI

Tech Stack: Next.js, Python, SQLite, OpenAI/Gemini API | **Origin:** Self-initiated / Personal Project

Team: Individual Project | **My Role:** Solo Developer

Project Summary

An AI-powered learning path generator that instantly designs structured curriculum milestones and study pathways tailored to a user's career track or technical goals.

My Impact & Contribution

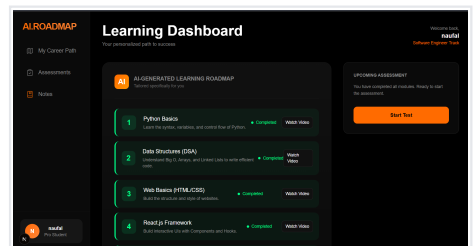
Architected the complete system from scratch, creating a responsive React-based node visualizer frontend, a Python API backend, and a local SQLite database to cache generated roadmaps.

Key Learnings

Gained deep experience in API latency management, crafting JSON-schema-compliant system prompts for deterministic LLM outputs, and managing tree-node hierarchies in client-side state.

Reference Links

GitHub: github.com/darrellrafa/Roadmap-ai



PROJECT #3: NutriBot

Tech Stack: Python, SQLite, Fine-tuned LLM | **Origin:** Coursework (Artificial Intelligence)

Team: Individual Project | **My Role:** *Solo Developer*

Project Summary

An AI chatbot assistant designed to simplify nutrition consulting, calorie tracking, and customized meal prep planning based on personalized user health profiles.

My Impact & Contribution

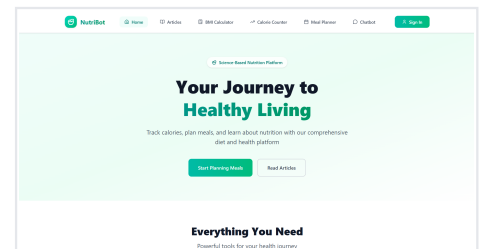
Set up database layers, wrapped REST API endpoints, and fine-tuned LLM hyperparameters to serve accurate and context-aware nutrition plans.

Key Learnings

Developed skills in LLM fine-tuning methodologies, structuring user health profile databases, and designing accessible, conversational UX flows for wellness tools.

Reference Links

GitHub: github.com/darrellrafa/Nutribot | Live: nutribot-testing.vercel.app



PROJECT #4: Design Anything

Tech Stack: Python, Flask, AI Image APIs, HTML/JS | **Origin:** Self-initiated Project

Team: Group Project (3 Members) | **My Role:** *Backend Developer & AI API Integrator*

Project Summary

A web platform integrating advanced image manipulation APIs for AI-driven photo edits like background removal, style transfers, and smart background replacements.

My Impact & Contribution

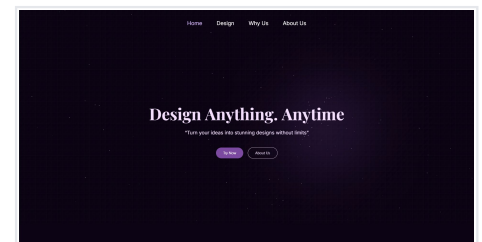
Designed the Python Flask backend server, integrating external AI model inference pipelines to handle and process image streams in real time.

Key Learnings

Gained depth in processing binary image data, managing caching for media assets, and resolving external API performance bottle-necks during heavy concurrent loads.

Reference Links

GitHub: github.com/darrellrafa/DesignEverything



PROJECT #5: Meet & Fight

Tech Stack: Flutter, Appwrite (Backend), Mobile UI | **Origin:** Self-initiated Project

Team: Group Project (3 Members) | **My Role:** *Lead Mobile Developer*

Project Summary

A cross-platform mobile application designed to connect local martial artists, enabling them to find and schedule sparring partners safely.

My Impact & Contribution

Crafted the complete mobile user interface in Flutter and connected user profiles, locations, and matchmaking databases using Appwrite database and real-time triggers.

Key Learnings

Mastered mobile state management (Provider/Bloc), custom interactive map components, and Appwrite real-time database queries.

Reference Links

GitHub: github.com/darrellrafa/might | Live: meetandfight.vercel.app

